

variance_inflation_factor from statsmodels.stats.outliers_influence – explanation

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
def calc_vif(X):
    #Calculating VIF
    vif = pd.DataFrame()
    vif["variables"] = X.columns
    vif["VIF"] = [variance_inflation_factor(X.values,i) for i in range(X.shape[1])]
    return(vif)
```

Here we defined a function to calculate the Variance Inflation Factor (VIF) for a given dataset X using the statsmodels library. VIF is a measure used to detect multicollinearity in regression analysis.

- The function is named calc_vif(X).
- It utilizes pandas to create a DataFrame to store the results.
- It iterates through the columns of the input data X to compute the VIF for each variable.
- VIF values exceeding 5 or 10 typically indicate high multicollinearity between an independent variable and others in the model.

Multicollinearity:

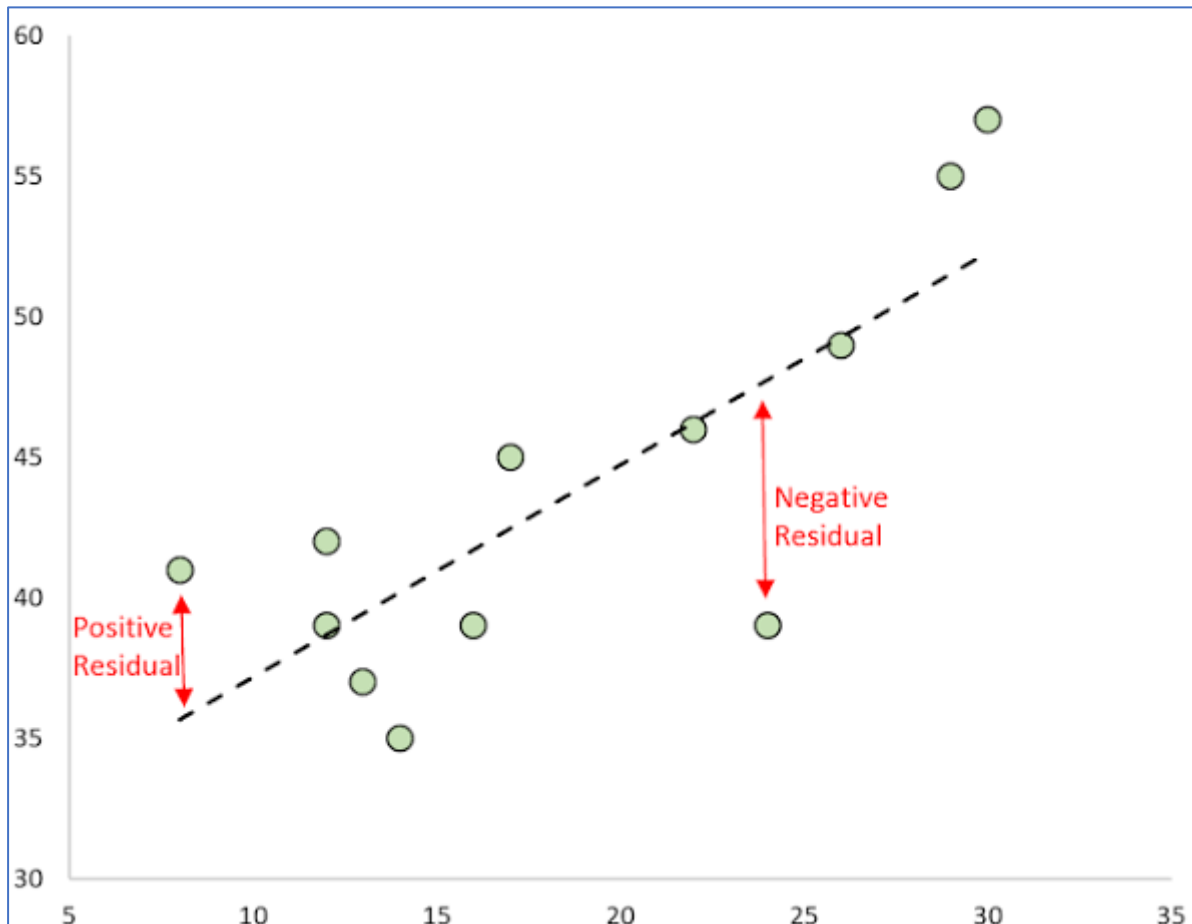
Multicollinearity occurs in regression analysis when two or more independent variables are highly correlated, making it difficult to isolate their individual effects on the dependent variable. While it does not reduce the model's overall predictive power, it inflates standard errors, making coefficient estimates unreliable and unstable.

Key Aspects of Multicollinearity

- Definition: A phenomenon where predictor variables in a model are linearly dependent or highly correlated.
- Types:
 - Perfect Multicollinearity: One predictor is an exact linear combination of others; the model cannot be computed.
 - Imperfect (Extreme) Multicollinearity: Predictors are highly correlated, causing inflated variance.
- Consequences: Coefficients become sensitive to minor changes in the model, p-values become unreliable, and interpreting individual variable impacts becomes impossible.
- Detection Methods:
 - [Variance Inflation Factor](#) (VIF): Measures how much the variance of an estimated regression coefficient is increased. A VIF > 5 or 10 indicates high multicollinearity.
 - Correlation Matrix: Identifies high correlations between pairs of variables.
 - Condition Number: A value > 30 suggests strong multicollinearity.
- Solutions:

- Remove Variables: Drop one of the highly correlated variables.
- Combine Variables: Create a new variable, such as an average or index, from the correlated ones.
- Regularization: Use techniques like Ridge or Lasso regression to stabilize coefficients.
- Increase Data: Sometimes, gathering more data reduces the strength of the correlation.

Residuals in regression:



Residuals in regression are the differences between observed dependent variable values (y) and predicted values ($\hat{y}$), representing the vertical distance from data points to the regression line ($e = y - \hat{y}$). They measure the prediction error, with sums of residuals totaling zero. Residual analysis checks model assumptions—linearity, homoscedasticity, and independence—via scatter plots. [🔗](#)

How to get error of particular model in regression:

To get the error of a particular model in regression, you need to calculate the residuals (the difference between the observed and predicted values) and then use a specific evaluation metric to summarize these residuals into a single value.

1. Calculate the Residuals

A residual (e_i) is the difference between an observed actual value (y_i) and the predicted value ($\hat{y}_i$) from your model for each data point.  Penn State University

$$e_i = y_i - \hat{y}_i$$

You will have one residual for every data point in your dataset. 

2. Choose an Evaluation Metric

Since inspecting thousands or millions of individual residuals is not practical, summary statistics are used to represent the model's overall performance. The most common metrics are:  Dataquest

- **Mean Absolute Error (MAE):** This is the average of the absolute differences between predictions and actual values. It is in the same units as the original data and is easy to interpret as the average prediction error magnitude.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Mean Squared Error (MSE):** This calculates the average of the *squared* differences. Squaring the errors penalizes larger errors more heavily than smaller ones, making it sensitive to outliers. The units are the square of the original target variable's units (e.g., "squared dollars"), which can be hard to interpret.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Root Mean Squared Error (RMSE):** This is the square root of the MSE and brings the error metric back into the same units as the original data, making it more interpretable than MSE.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **R-squared (R^2):** This metric provides a relative measure of how well the model fits the data, indicating the proportion of the variance in the dependent variable that is explained by the model. It ranges from 0 to 1 (or 0% to 100%), with higher values indicating a better fit.  Statistics By Jim +4

[Homoscedasticity](#) is generally better for linear regression, as it satisfies the ordinary least squares (OLS) assumption of constant error variance, leading to more reliable, efficient, and unbiased, precise predictions. Conversely, [heteroscedasticity](#) (unequal variance) violates these assumptions, often resulting in skewed, unreliable standard errors and misleading, invalid p-values.

Homoscedasticity vs. Heteroscedasticity Comparison

- **Homoscedasticity (Preferred):** Residuals are spread evenly across all levels of the independent variable, maintaining constant variance. It is crucial for accurate and reliable statistical tests.
- **Heteroscedasticity (Problematic):** The spread of residuals changes (e.g., fan/cone shape) depending on the predictor values. It indicates that the model is less accurate at certain data ranges.

Key Takeaways:

- **Why Homoscedasticity Wins:** It allows for trustworthy inference and reliable confidence intervals.
- **Fixing Heteroscedasticity:** If present, one can use transformations (e.g., logarithms), weighted least squares, or heteroscedasticity-consistent standard errors (HCSE) to correct the model.
- **Reality Check:** While homoscedasticity is desired, real-world data frequently exhibits heteroscedasticity, requiring specialized techniques to handle it.